SpaceX Falcon 9
First-Stage Landing Prediction

IBM Data Science Capstone Project

Final presentation | Data collection, analysis, interactive visualization, and machine learning

Dataset period: 2010-2020

GitHub Repository URL

Public project repository

https://github.com/TristinWen/spacex-falcon9-landing-prediction

Repository evidence

Seven completed notebooks • Dash source and screenshots • Folium map • Final PPTX/PDF • Final DOCX/PDF report

EXECUTIVE SUMMARY

The analysis identifies where and when Falcon 9 landings succeed

A reusable first stage lowers launch cost, so predicting landing success supports competitive bid estimates.

90

Falcon 9 launches

87.8%

successful landings in prepared labels

83.3%

test accuracy across four models

41.7%

share of all successful launches at KSC LC-39A

Key implication

Operational history and launch context provide useful predictive signal, while launch site, orbit, payload, and booster configuration explain meaningful differences in outcomes.

Predicting first-stage landing success supports launch-cost estimates

Business context

SpaceX can offer lower launch prices partly because Falcon 9 first stages are designed for recovery and reuse.

Analytical problem

Determine which mission characteristics are associated with a successful landing and compare classification models that predict the outcome.

Target variable: Class = 1 for successful landing; Class = 0 for unsuccessful landing.

METHOD

The project combines seven analytical stages

01 API collection → 02 Web scraping → 03 Wrangling → 04 EDA + SQL → 05 Folium + Dash → 06 Feature engineering → 07 Model selection

Evidence was retained in executed notebooks, dashboard screenshots, and an interactive Folium HTML map.

SpaceX API data collection produces 90 Falcon 9 launch records

Request and parse launch JSON

- Request past launches from the SpaceX REST API
- Normalize JSON into a Pandas DataFrame
- Resolve rocket, payload, launchpad, and core IDs
- Retain booster, payload, orbit, site, outcome, reuse, and coordinates

90

Falcon 9 launch records

17 analytical fields

Web Scraping extracts 121 historical launch-table rows

BeautifulSoup parsing workflow

- Request the Wikipedia Falcon 9 launch snapshot
- Extract HTML table headers
- Parse flight, date, booster, site, payload, orbit, customer, outcome, and landing
- Convert the launch dictionary to a Pandas DataFrame

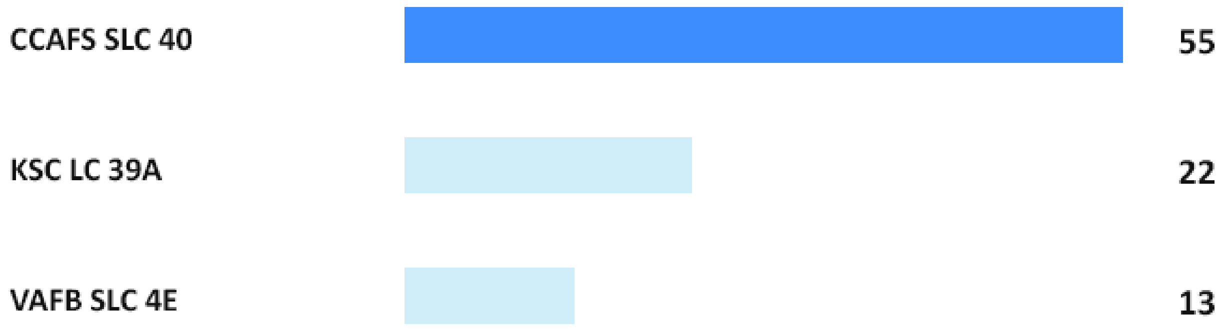
121

parsed launch rows

11 output columns

Data Wrangling creates a consistent landing-success target

Launches by site

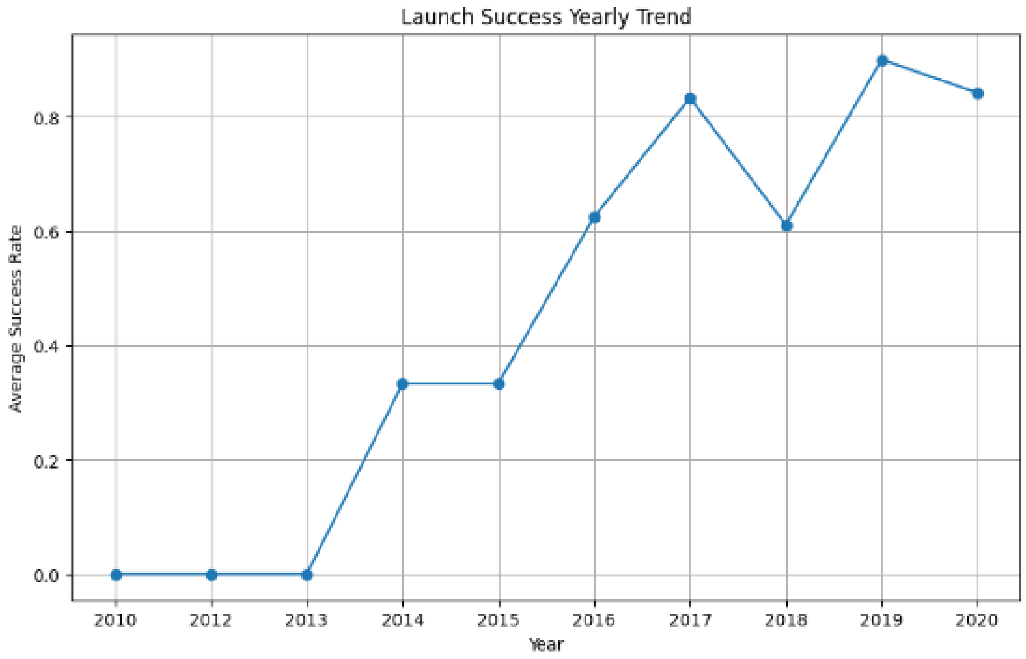


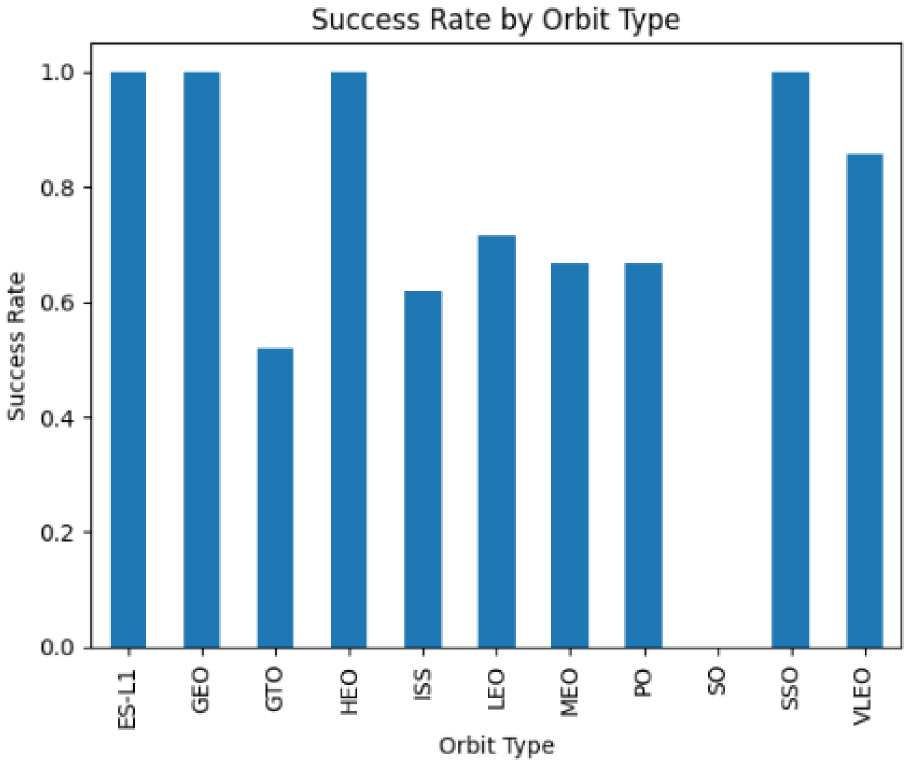
Wrangling steps

- Replace missing PayloadMass with the mean
- Retain LandingPad nulls as meaningful “no pad” values
- Map unsuccessful outcomes to Class = 0
- Map remaining outcomes to Class = 1

Result: 90 rows, no missing PayloadMass values, and an 87.8% success share.

EDA with Visualization shows success improving over time





Yearly Launch Success Trend

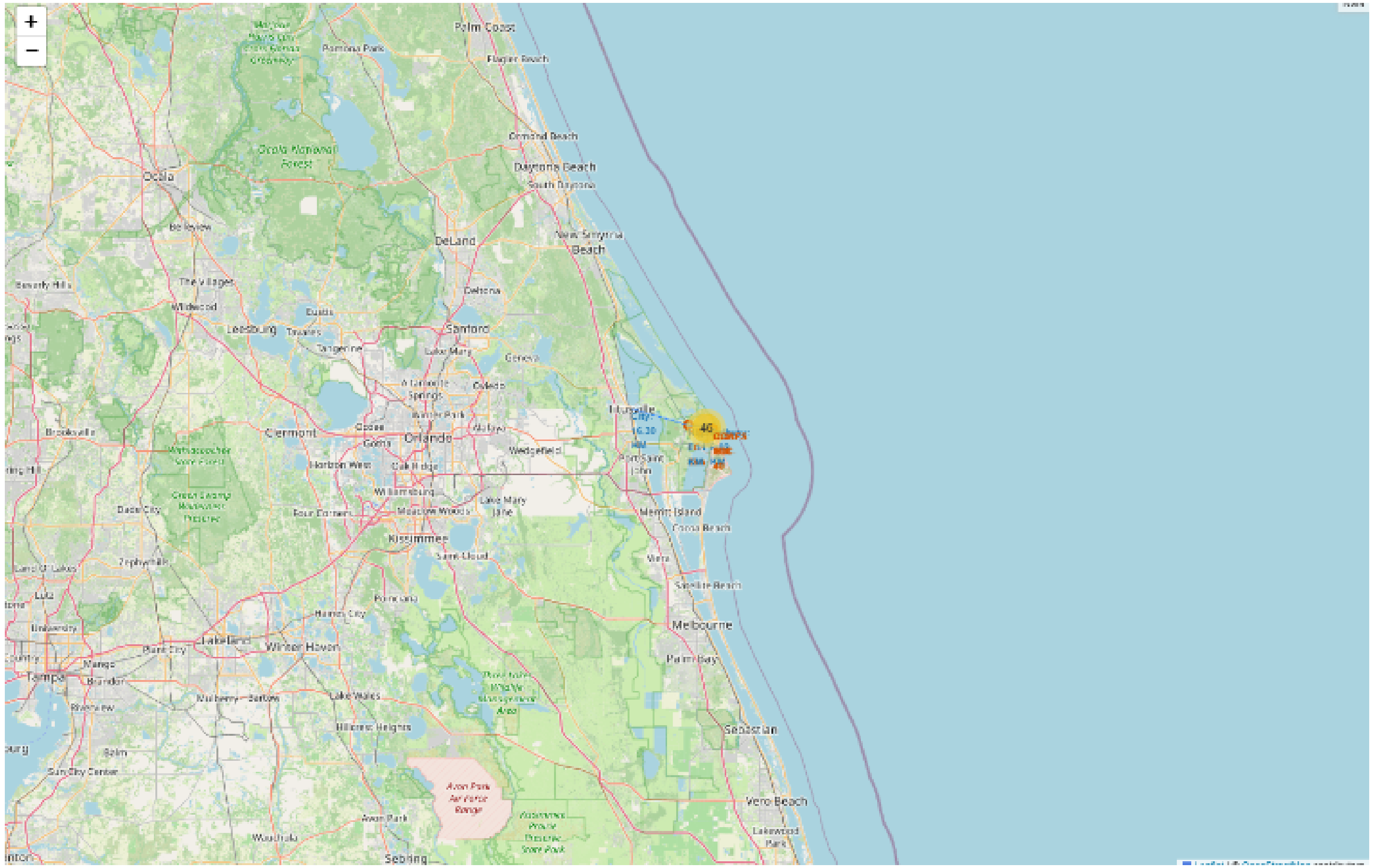
Success Rate by Orbit Type

EDA with SQL confirms milestones and payload patterns

NASA CRS payload	45,596 kg
Avg. F9 v1.1 payload	2,928.4 kg
First ground-pad success	22 Dec 2015
Successful missions	100*
Drone-ship successes, 2010-2017	5

SQL Tasks 1-10 completed; *source records include “Success” and “Success ” as separate values.

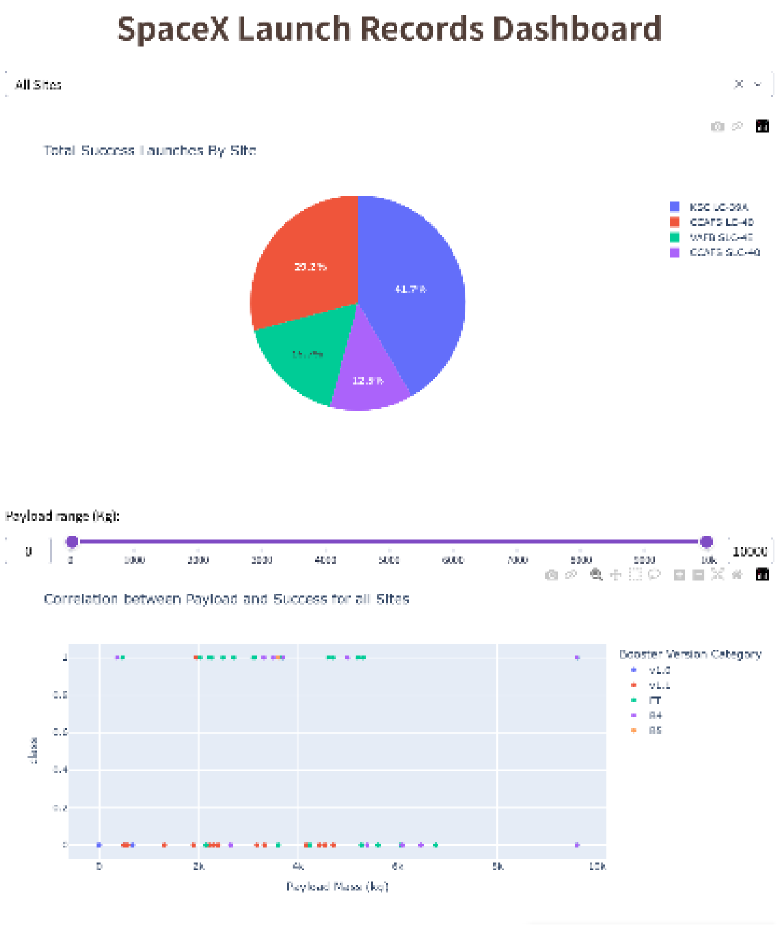
Interactive Visual Analytics with Folium maps launch-site proximity



Required Folium evidence

- All launch sites marked
- Success and failure markers clustered
- Distance to coastline calculated
- Lines to city, railway, and highway proximities

Plotly Dash: All-Sites Success Pie Chart

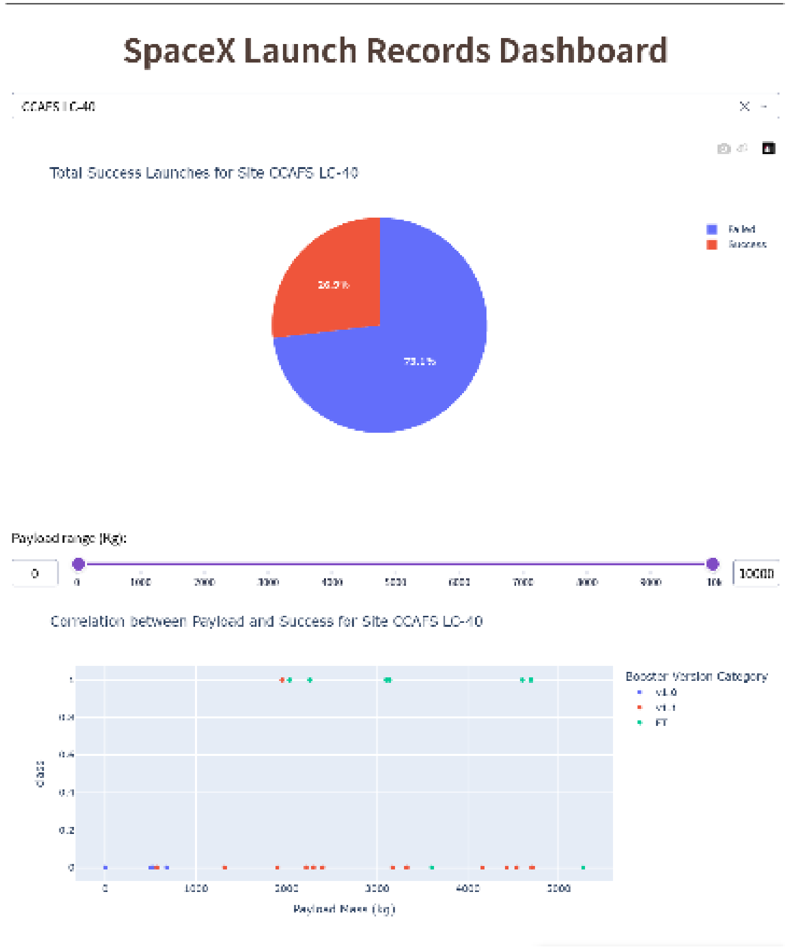


41.7%

KSC LC-39A has the largest share of successful launches

Evidence: launch-site dropdown, success pie chart, payload range slider, and payload-versus-outcome scatter plot.

Plotly Dash: Site Filter and Payload Scatter Plot



26.9%

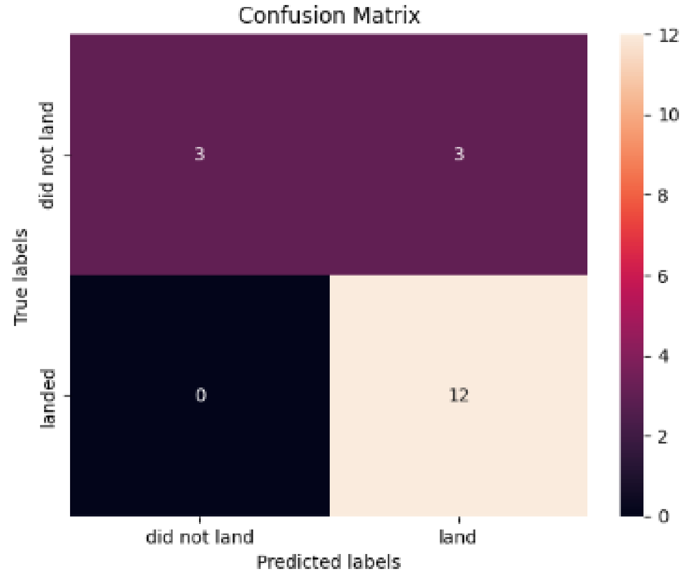
CCAFS LC-40 successful launches

73.1% failed

Selecting a site updates both the success/failure pie chart and payload-versus-launch-outcome scatter plot.

Predictive Analysis: all four models reach 83.3% test accuracy

Model	CV accuracy	Test accuracy
Logistic regression	84.6%	83.3%
Support vector machine	84.8%	83.3%
Decision tree	92.9%	83.3%
K-nearest neighbors	84.8%	83.3%



Models evaluated: Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors.

Decision tree leads cross-validation, while identical test scores reflect the small 18-case test set.

Results and Findings: landing success reflects operational maturity

01	**Performance improves over time**	Later flights show stronger landing outcomes as operations mature.
02	**Site and orbit matter**	KSC leads successful launches; outcome rates differ across sites and orbit profiles.
03	**Reusable hardware features carry signal**	Grid fins, legs, reuse history, block, orbit, and payload contribute to prediction.
04	**Models require broader validation**	83.3% test accuracy is promising, though the test set contains only 18 launches.

Submission package maps directly to the capstone rubric

Included

- API collection notebook
- Web scraping notebook
- Data wrangling notebook
- EDA visualization notebook
- SQL notebook
- Folium notebook + HTML map
- Dash application + screenshots
- Machine learning notebook

Public GitHub repository

All completed notebooks, dashboard code, screenshots, map, presentation, and report are available in the repository.

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IBM CAPSTONE AUTOMATED-GRADING INDEX

Rubric Evidence Index